

QR Factorization and Least Squares

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QR Factorization

Orthogonal Vectors

From Section 1.ii of the Hefferon book: two vectors from $\mathbb{R}^n$ are called *orthogonal* (or *perpendicular*) if their dot product is zero. It makes sense to identify $\vec{x} \cdot \vec{y}$ and $\vec{x}^\top \vec{y}$.

The book also writes $|\vec{x}| := \sqrt{\vec{x} \cdot \vec{x}}$ for the (Euclidean) length of $\vec{x}$. More often this is written as $\|\vec{x}\|$ or $\|\vec{x}\|_2$ (“2-norm” or “ L_2 -norm” or “ ℓ_2 -norm”).

From Section 3.vi:

2.2 Theorem If the vectors in a set $\{\vec{v}_1, \dots, \vec{v}_k\} \subset \mathbb{R}^n$ are pairwise orthogonal and nonzero then that set is linearly independent.

Proof Consider $c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_k\vec{v}_k = \vec{0}$. For $i \in \{1, \dots, k\}$, taking the dot product of $\vec{v}_i$ with both sides of the equation $\vec{v}_i \cdot (c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_k\vec{v}_k) = \vec{v}_i \cdot \vec{0}$, which gives $c_i \cdot (\vec{v}_i \cdot \vec{v}_i) = 0$, shows that $c_i = 0$ since $\vec{v}_i \neq \vec{0}$. QED

So these vectors form a basis of the vector space they span.

Orthogonal Matrices

Recall $(AB)^\top = B^\top A^\top$ and (if it exists) $(AB)^{-1} = B^{-1}A^{-1}$.

A square matrix is called *orthogonal* (or *orthonormal*) if its columns have length 1 and are pairwise orthogonal. Equivalently, $Q \in \mathbb{R}^{n \times n}$ is orthogonal if and only if $Q^\top = Q^{-1}$, i.e., $Q^\top Q = I$:

$$\begin{pmatrix} \text{---} \vec{q}_1^\top \text{---} \\ \text{---} \vec{q}_2^\top \text{---} \\ \vdots \\ \text{---} \vec{q}_n^\top \text{---} \\ Q^\top \end{pmatrix} \begin{pmatrix} \left| \begin{array}{c} \vec{q}_1 \\ \vec{q}_2 \\ \vdots \\ \vec{q}_n \end{array} \right| \\ Q \end{pmatrix} = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \\ I \end{pmatrix}$$

Multiplication by an orthogonal matrix preserves the length of vectors:

$$|\vec{Q}\vec{x}|^2 = (\vec{Q}\vec{x})^\top \vec{Q}\vec{x} = \vec{x}^\top Q^\top Q \vec{x} = \vec{x}^\top \vec{x} = |\vec{x}|^2$$

Similarly, angles are preserved:

$$(\vec{Q}\vec{x}) \cdot (\vec{Q}\vec{y}) = (\vec{Q}\vec{x})^\top \vec{Q}\vec{y} = \vec{x}^\top Q^\top Q \vec{y} = \vec{x}^\top \vec{y} = \vec{x} \cdot \vec{y}$$

Geometrically, multiplication by an orthogonal matrix is a rotation/reflection.

We will consider non-square matrices in the following. In particular also $m \times n$ matrices Q (with $m \geq n$) whose columns have length 1 and are pairwise orthogonal:

$$\begin{pmatrix} \text{---} \vec{q}_1^\top \text{---} \\ \vdots \\ \text{---} \vec{q}_n^\top \text{---} \end{pmatrix} \begin{pmatrix} \left| \begin{array}{c} \\ \vec{q}_1 \\ \end{array} \right| & \cdots & \left| \begin{array}{c} \\ \vec{q}_n \\ \end{array} \right| \end{pmatrix} = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$$

$$\boxed{Q^\top} \quad \boxed{Q} = \boxed{I}$$

Gram-Schmidt Orthogonalization

Let $\vec{a}_1, \dots, \vec{a}_n \in \mathbb{R}^m$ be linearly independent vectors (so $m \geq n$). In Section 3.vi we have seen the Gram-Schmidt orthogonalization process:

$$\vec{q}_1 := \vec{a}_1$$

$$\vec{q}_2 := \vec{a}_2 - \frac{\vec{q}_1^\top \vec{a}_2}{\vec{q}_1^\top \vec{q}_1} \vec{q}_1$$

$$\vdots$$

$$\vec{q}_n := \vec{a}_n - \frac{\vec{q}_1^\top \vec{a}_n}{\vec{q}_1^\top \vec{q}_1} \vec{q}_1 - \dots - \frac{\vec{q}_{n-1}^\top \vec{a}_n}{\vec{q}_{n-1}^\top \vec{q}_{n-1}} \vec{q}_{n-1}$$

Then $\langle \vec{q}_1, \dots, \vec{q}_n \rangle$ is an orthogonal basis for the span of $\vec{a}_1, \dots, \vec{a}_n$.

It is natural to normalize the $\vec{q}_i$ on the go:

$$\begin{aligned}\vec{q}_1 &:= \frac{\vec{a}_1}{|\vec{a}_1|} \\ \vec{q}_2 &:= \frac{\vec{a}_2 - (\vec{q}_1^\top \vec{a}_2) \vec{q}_1}{|\vec{a}_2 - (\vec{q}_1^\top \vec{a}_2) \vec{q}_1|} \\ &\vdots \\ \vec{q}_n &:= \frac{\vec{a}_n - (\vec{q}_1^\top \vec{a}_n) \vec{q}_1 - \cdots - (\vec{q}_{n-1}^\top \vec{a}_n) \vec{q}_{n-1}}{|\vec{a}_n - (\vec{q}_1^\top \vec{a}_n) \vec{q}_1 - \cdots - (\vec{q}_{n-1}^\top \vec{a}_n) \vec{q}_{n-1}|}\end{aligned}$$

Defining

$$r_{ij} := \vec{q}_i^\top \vec{a}_j \quad (\text{for } i < j) \quad \text{and} \quad r_{ii} := |\vec{a}_i - (\vec{q}_1^\top \vec{a}_i) \vec{q}_1 - \cdots - (\vec{q}_{i-1}^\top \vec{a}_i) \vec{q}_{i-1}|$$

we have:

$$\begin{aligned}r_{11} \vec{q}_1 &= \vec{a}_1 \\ r_{22} \vec{q}_2 &= \vec{a}_2 - r_{12} \vec{q}_1 \\ &\vdots \\ r_{nn} \vec{q}_n &= \vec{a}_n - r_{1n} \vec{q}_1 - \cdots - r_{n-1,n} \vec{q}_{n-1}\end{aligned}$$

Putting the $\vec{a}_i$ on the left-hand side

$$\vec{a}_1 = r_{11}\vec{q}_1$$

$$\vec{a}_2 = r_{12}\vec{q}_1 + r_{22}\vec{q}_2$$

$$\vdots$$

$$\vec{a}_n = r_{1n}\vec{q}_1 + r_{2n}\vec{q}_2 + \cdots + r_{n-1,n}\vec{q}_{n-1} + r_{nn}\vec{q}_n$$

we see that we have obtained a *QR factorization* of $A := (\vec{a}_1 \ \cdots \ \vec{a}_n)$:

$$\left(\begin{array}{c|c|c|c} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{array} \right) = \left(\begin{array}{c|c|c|c} \vec{q}_1 & \vec{q}_2 & \cdots & \vec{q}_n \end{array} \right) \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ & r_{22} & \cdots & r_{2n} \\ & & \ddots & \vdots \\ & & & r_{nn} \end{pmatrix}$$

A

=

Q

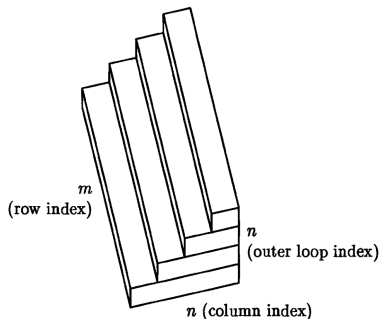
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R

This is also called a *reduced/thin* QR factorization. A *full* QR factorization has additional columns in Q to make it square and corresponding zeros in R.

Algorithm 1: QR Factorization via Classical Gram-Schmidt (unstable)

```
for j := 1 to n
   $\vec{q}_j := \vec{a}_j$ 
  for i := 1 to j - 1
     $r_{ij} := \vec{q}_i^\top \vec{a}_j$ 
     $\vec{q}_j := \vec{q}_j - r_{ij} \vec{q}_i$ 
   $r_{jj} := |\vec{q}_j|$ 
   $\vec{q}_j := \vec{q}_j / r_{jj}$ 
```



Each iteration of the innermost loop involves roughly $2m$ (scalar) multiplications and $2m$ additions/subtractions.

Overall about $4 \cdot (\frac{1}{2} mn^2) = 2mn^2$ floating-point operations.

Algorithm 2: QR Factorization via *Modified* Gram-Schmidt (stable)

```
for i := 1 to n
   $\vec{v}_i := \vec{a}_i$ 
for i := 1 to n
   $r_{ii} := |\vec{v}_i|$ 
   $\vec{q}_i := \vec{v}_i / r_{ii}$ 
  for j := i + 1 to n
     $r_{ij} := \vec{q}_i^\top \vec{v}_j$ 
     $\vec{v}_j := \vec{v}_j - r_{ij} \vec{q}_i$ 
```

Here, each $\vec{v}_i$ starts out as $\vec{a}_i$. From $\vec{v}_i$ the $\vec{q}_j$ -components for $j < i$ are removed as soon as they become available. So the computation of the r_{ij} is not “disturbed” by the presence of already removed components.

This variant is mathematically equivalent, but turns out to be numerically stable.

The number of flops stays the same.

Gram-Schmidt can be viewed as [triangular orthogonalization](#):

$$A \underbrace{R_1 R_2 \cdots R_n}_{R^{-1}} = Q \quad (\text{reduced QR factorization})$$

Alternative methods proceed via [orthogonal triangularization](#):

$$\underbrace{Q_n \cdots Q_2 Q_1}_{Q^T = Q^{-1}} A = R \quad (\text{full QR factorization})$$

$$\begin{pmatrix} \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \end{pmatrix} \xrightarrow{Q_1} \begin{pmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \end{pmatrix} \xrightarrow{Q_2} \begin{pmatrix} \times & \times & \times \\ & \times & \times \\ & 0 & \times \\ & 0 & \times \\ & 0 & \times \end{pmatrix} \xrightarrow{Q_3} \begin{pmatrix} \times & \times & \times \\ & \times & \times \\ & & \times \\ & & 0 \\ & & 0 \end{pmatrix}$$

A
 $Q_1 A$
 $Q_2 Q_1 A$
 $Q_3 Q_2 Q_1 A$

Such methods use *Householder reflections* or *Givens rotations*. They are numerically stable (as multiplication with orthogonal matrices tends to be). Operation count (Householder and $m = n$): about $\frac{4}{3}n^3$.

Least Squares

Solving Linear Systems via QR Factorization

Consider a **square** system $A\vec{x} = \vec{b}$ with A nonsingular, and a QR factorization $A = QR$. Then:

$$\begin{aligned} QR\vec{x} &= \vec{b} \\ \iff R\vec{x} &= Q^T\vec{b} \end{aligned}$$

So we can solve the system by first multiplying with Q^T and then backward substitution.

The overall cost is dominated by the factorization (e.g., Householder $\frac{4}{3}n^3$ vs. Gauss $\frac{2}{3}n^3$).

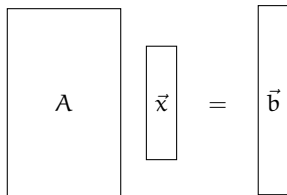
The advantage is improved numerical stability.

Don't invert A . Instead (QR-)factorize A !

Least Squares

Consider $A\vec{x} = \vec{b}$ with $A \in \mathbb{R}^{m \times n}$:

more equations than unknowns (“overdetermined”)


$$A \vec{x} = \vec{b}$$

We assume that A has full column rank n , i.e., the columns are linearly independent. In general the system cannot be solved exactly.

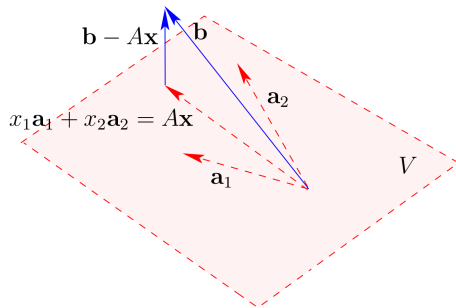
The *least squares* problem is to find $\vec{x}$ such that $\|\vec{b} - A\vec{x}\|_2$ is minimized.

Normal Equations

$$A \vec{x} \approx \vec{b}$$

$$\begin{pmatrix} | & | \\ \vec{a}_1 & \vec{a}_2 \\ | & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \approx \begin{pmatrix} | \\ \vec{b} \\ | \end{pmatrix}$$

$$x_1 \vec{a}_1 + x_2 \vec{a}_2 \approx \vec{b}$$



By the Pythagorean theorem, $\vec{x}$ should be chosen so that the “residual” $\vec{b} - A\vec{x}$ is orthogonal to the (hyper-)plane spanned by the columns of A . Equivalently:

$$\begin{aligned} A^\top (\vec{b} - A\vec{x}) &= \vec{0} \\ \iff A^\top A\vec{x} &= A^\top \vec{b} \end{aligned}$$

Uniqueness

We just saw the *normal equations* $A^T A \vec{x} = A^T \vec{b}$. Can this system be solved uniquely?

Lemma If $A \in \mathbb{R}^{m \times n}$ has full column rank n , then $A^T A$ is nonsingular.

Proof We show that the nullspace of $A^T A$ is trivial. Let $\vec{x} \in \mathbb{R}^n$ with $A^T A \vec{x} = \vec{0}$. Then $\|A\vec{x}\|^2 = (A\vec{x}) \cdot (A\vec{x}) = \vec{x}^T A^T A \vec{x} = 0$, so $A\vec{x} = \vec{0}$. Since the columns of A are linearly independent, it follows $\vec{x} = \vec{0}$. QED

It follows that the normal equations can be solved uniquely, and so there is a unique solution to the least squares problem. Defining the *pseudoinverse* $A^+ := (A^T A)^{-1} A^T$ the solution is $\vec{x} = A^+ \vec{b}$.

As ever, inverting $A^T A$ is not a good algorithm. One solution method takes advantage of the fact that $A^T A$ is *positive definite*, i.e., it is symmetric and $\vec{z}^T A^T A \vec{z} > 0$ holds for all nonzero $\vec{z}$. Such matrices have a *Cholesky factorization* $R^T R$ where R is upper-triangular. It can be computed relatively fast. Given that factorization, one can then solve for $\vec{x}$ similarly as discussed for LU. Numerical stability is a concern though.

Least Squares via QR Factorization

Let $A = QR$ be a reduced QR factorization. Then

$$\begin{aligned} A^T A \vec{x} &= A^T \vec{b} \\ \iff R^T \underbrace{Q^T Q}_I R \vec{x} &= R^T Q^T \vec{b} \end{aligned}$$

Since A has full column rank, the matrix R is nonsingular. Thus, multiplying with the inverse of R^T gives the system:

$$R \vec{x} = Q^T \vec{b}$$

Algorithm 3: Least Squares via QR Factorization

Compute the reduced QR factorization $A = QR$

Compute the vector $Q^T \vec{b}$

Solve the upper-triangular system $R \vec{x} = Q^T \vec{b}$ via back substitution

Linear Regression

Progression of world record times for the men's mile race:

<i>year</i>	1870	1880	1890	1900	1910	1920	1930	1940	1950
<i>secs</i>	268.8	264.5	258.4	255.6	255.6	252.6	250.4	246.4	241.4

When would the 4-minute mark fall?

$$c + 1870m = 268.8$$

$$c + 1880m = 264.5$$

$$\vdots$$

$$c + 1950m = 241.4$$

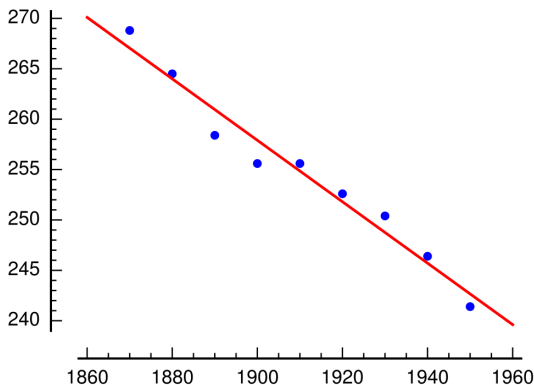
$$A = \begin{pmatrix} 1 & 1870 \\ 1 & 1880 \\ \vdots & \vdots \\ 1 & 1950 \end{pmatrix} \quad \vec{x} = \begin{pmatrix} c \\ m \end{pmatrix} \quad \vec{b} = \begin{pmatrix} 268.8 \\ 264.5 \\ \vdots \\ 241.4 \end{pmatrix}$$

We want to find $\vec{x} = \begin{pmatrix} c \\ m \end{pmatrix}$ that minimizes $|A\vec{x} - \vec{b}|$.

Solving the normal equations $A^T A \vec{x} = A^T \vec{b}$ yields:

$$\vec{x} = \begin{pmatrix} c \\ m \end{pmatrix} \approx \begin{pmatrix} 837 \\ -0.305 \end{pmatrix}$$

This gives the *line of best fit*:



Concerning the 4-minute mark, we would predict 1958.73. The actual date was 6 May 1954 (Roger Bannister, Iffley Road track, Oxford).